



Application Note

EventAnalysisTool

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Introduction

EventAnalysisTool is a very useful tool to assist with developing and debugging a SetUp script. After running a SetUp script, so its output is present in the Matlab workspace, just enter *EventAnalysisTool* at the Matlab command prompt and you will see a visual display of all events in the event sequence. Clicking on any Event will display the contents of structures indexed by that Event, making it easy to find and fix errors in indexing logic as well as helping you understand what the system will actually be doing as you step through each event in the event sequence. In addition, debugging is very frustrating, so debugging capability is added into the EventAnalysisTool. An overview with few example scripts is listed here and a demonstration video is provided on Verasonics website: <http://verasonics.com/tools-demonstration/>

Getting Start

Starting from software release 4.2, the initializeOnly feature and VsUpdate will not be applied by default. The initializeOnly mode can be executed by clicking the "initializeOnly mode" push button or type "EventAnalysisTool i" in the command window for more information (slower execution). The following message will be displayed while parsing the script.

Parsing the script using initializeOnly. Please wait...

Debugging strategy

Because the event sequence contains different indices, it's easy to have indexing errors. The EventAnalysisTool checks possible errors as follows (some critical bugs are listed in red):

1. Trans, Resource, TW, TX and Event structures are required as minimum.
2. Check required fields of TX, Receive, and Process objects.
3. **All fields of each event should not be empty.**
4. Usually, each TX, Receive, Recon, Process, and SeqControl should be used in the event sequence. However, it's legitimate to use less in the event sequence, so only a warning message will be provided. For instance, some SeqControl objects are predefined for testing, not all SeqControl objects are necessary.
5. It's not legitimate if the index in the event sequence is larger than the number of associated structure. For instance, Receive(1000) is used in the event sequence but only 800 Receive objects are defined.
6. Identical Receive object should not be used in multiple events.
7. Each Receive should be unique, in terms of endDepth, bufnum, framenum, acqNum with mode 0; otherwise the RcvData will be overwritten by later rev event.

8. The endDepth, bufnum, and framenum of all Receive objects used in one Recon should be identical.
9. Process.Parameters can be an empty cell but not an empty array
10. SeqControl.command can't be empty
11. returnToMatlab is required to prevent an endless loop, expect for the fact that jump back to the first event will add returnToMatlab automatically.
12. RcvData may only contain few frames for 'one-shot' sequence if the script doesn't jump without sync, triggerIn, or waitForProcessing. It's caused by the fact that the software sequencer reaches to the end of event earlier than the hardware sequencer.
13. The EventAnalysisTool estimates the interval between two events with timeToNextAcq SeqControl. If initializeOnly is applied, the size and transfer rate of each DMA will be calculated. **The next DMA won't be initialized before the current one is completed! Note: the overhead between events is an estimated value, 4 us, so the estimation here can only be used as the reference.**
14. **The index of transferToHost (DMA) needs to be unique in the event sequences**, i.e., the index of transferToHost can't be reused, so the transferToHost can't be predefined before the event sequence.
15. All bufnum and framenum of one transferToHost must be identical.

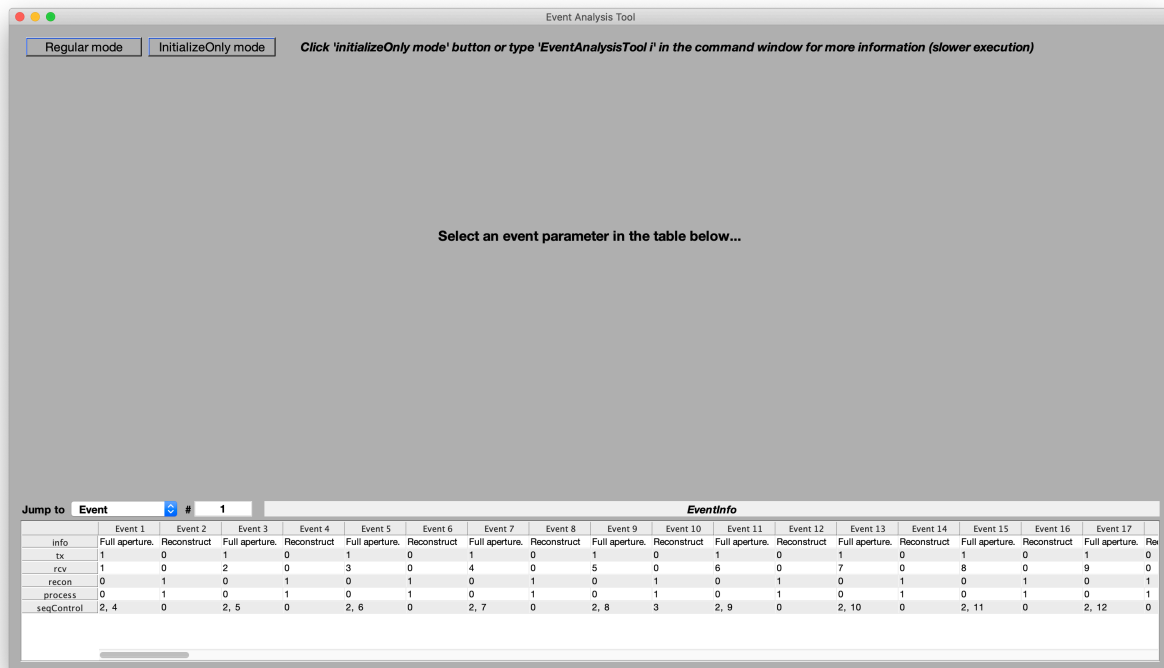
In the EventAnalysisTol/Examples subfolder, six error scripts are provided to illustrate how to use EventAnalysisTool to find bugs in the script. The corresponding corrected scripts are also provided.

- 1) ErrorScript1 - Warning message if any of event has an empty field
- 2) ErrorScript2 - Each TX and Receive should be used
- 3) ErrorScript3 - Consistency of frameNum or BufferNum in Recon/ReconInfo and transferToHost
- 4) ErrorScript4 - A bug in the callback! The endDepth of all receive used for one Recon should be identical
- 5) ErrorScript5 - Each Receive should not have the same bufnum, framenum, acqNum, and mode; otherwise the data will be overwritten to the same memory location.
- 6) ErrorScript6 - transferToHost must be unique in the event sequence.

Interface introduction

1. “Jump to” a certain event

If no error is detected, the interface of EventAnalysisTool is shown below:



The ‘Regular mode’ button is equivalent to the EventAnalysisTool without initializeOnly feature. If the user would like to execute initialization process, please click ‘initializeOnly mode’ button or type EventAnalysisTool with any argument in the command window.

The “Jump to” feature provides the flexibility for the user to select desired index of Event, tx, rcv, recon, process, and seqControl. If too many events are created in the SetUp script, it takes time for the user to scroll the event table to the correct position for correct index number. To resolve the issue, the user can use “Jump to” feature to jump to the desired event number or any index, like tx(2), rcv(100), or recon(3). **Note:** this feature may not work as expected because the screen size and resolution are taken into account. It may jump to the structure above or below the desired one. If it happens, please jump to event number instead or modify the EventAnalysisTool accordingly. The user can even type a keyword of the SeqControl.command, like TTNA, DMA, sync, jump, or trigger, etc..., and the table should jump to correct event accordingly. Below is an example of “Jump To” feature.

After running ErrorScript3, EventAnalysisTool will discover two critical errors:

Event Analysis Tool

NOTE: Window Can Be Maximized

Reload Workspace

Error!! Receive.frameenum across Receive 1 2 3 4 5... must be identical for Recon 1

Error!! Receive.frameenum across Receive 1 2 3 4 5... must be identical for SeqControl 5

Jump to Event # 1

	Event 1	Event 2	Event 3	Event 4	Event 5	Event 6	Event 7	Event 8	Event 9	Event 10	Event 11	Event 12	Event 13	Event 14
Info	Full aperture	Full aperture	Full aperture	Full aperture	Full aperture	Full aperture	Full aperture	Full aperture	Full aperture	Full aperture	Full aperture	Full aperture	Full aperture	Full aperture
tx	1	2	3	4	5	6	7	8	9	10	11	12	13	14
rcv	1	2	3	4	5	6	7	8	9	10	11	12	13	14
recon	0	0	0	0	0	0	0	0	0	0	0	0	0	0
process	0	0	0	0	0	0	0	0	0	0	0	0	0	0
seqControl	2	2	2	2	2	2	2	2	2	2	2	2	2	2

The user can select recon and keep the value of # as 1. However, the user need to click the edit box of “#” and then click enter to execute the callback to display the recon 1:

Event Analysis Tool

NOTE: Window Can Be Maximized

Reload Workspace

Reconstruction Object	Entry/Value
Recon(1).senscutoff	0.6
Recon(1).pdatanum	1
Recon(1).rcvBufFrame	1
Recon(1).IntBufDest	[1, 1]
Recon(1).ImgBufDest	[1, 1]
Recon(1).RIINums	210x1 double
Recon(1).numchannels	128

Source and Destination for Image Reconstruction			
Source	Destination	Destination	Destination
	RcvBuffer	InterBuffer	ImageBuffer
buffer num	1	1	1
frame num	Impossible!!	1	1

ReconInfo Structures		Source				Destination		
	mode	txnum	rcvnum	bufnum	framenum	acqNum	pagenum	regionnum
RI 001	replaceIQ	1	1	1	1	1	1	1
RI 002	replaceIQ	2	2	1	1	2	2	1
RI 003	replaceIQ	3	3	1	1	3	3	1
RI 004	replaceIQ	4	4	1	1	4	4	1
RI 005	replaceIQ	5	5	1	1	5	5	1
RI 006	replaceIQ	6	6	1	1	6	6	1
RI 007	replaceIQ	7	7	1	1	7	7	1
RI 008	replaceIQ	1	8	1	2	1	8	1
RI 009	replaceIQ	2	9	1	2	2	9	1
RI 010	replaceIQ	3	10	1	2	3	10	1
RI 011	replaceIQ	4	11	1	2	4	11	1

Error!! The framenum is not consistent!

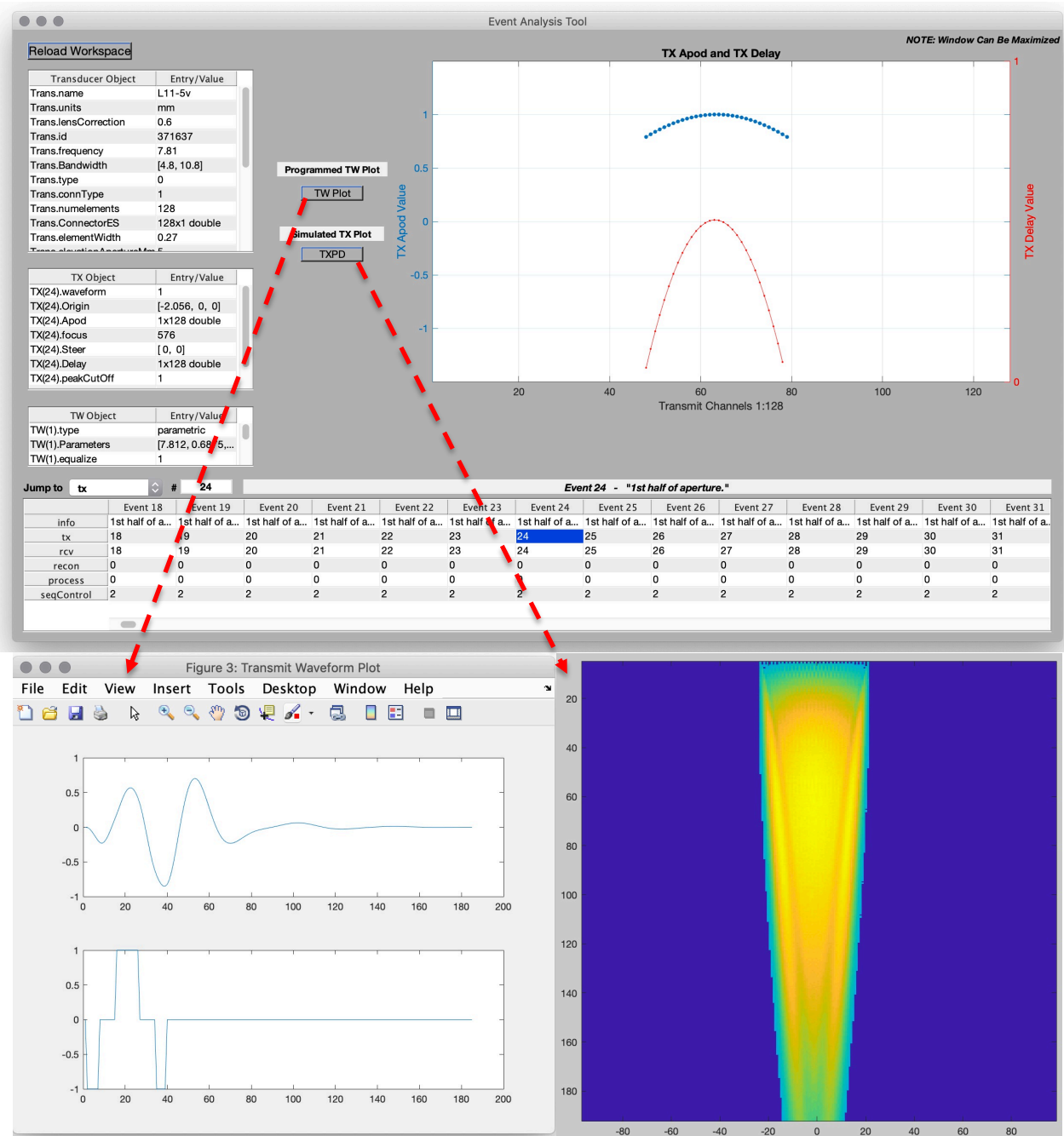
Jump to recon # 1

Event 211 - "recon and process"

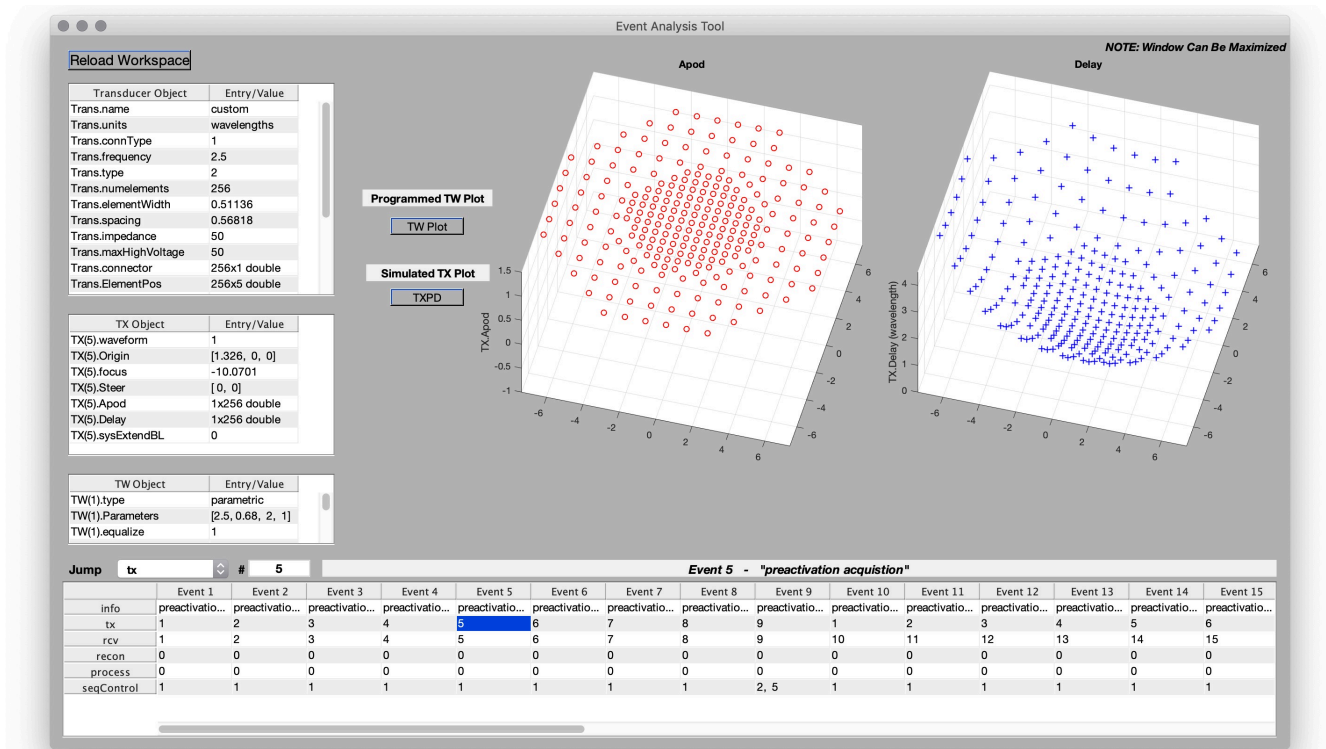
	Event 199	Event 200	Event 201	Event 202	Event 203	Event 204	Event 205	Event 206	Event 207	Event 208	Event 209	Event 210	Event 211	Event 212
Info	Full aperture	Full aperture	Full aperture	Full aperture	Full aperture	Full aperture	Full aperture	Full aperture	Full aperture	Full aperture	Full aperture	Full aperture	recon and pr...	Jump back
tx	4	5	6	7	1	2	3	4	5	6	7	0	0	
rcv	99	200	201	202	203	204	205	206	207	208	209	210	0	
recon	0	0	0	0	0	0	0	0	0	0	0	1	0	
process	0	0	0	0	0	0	0	0	0	0	0	1	0	
seqControl	2	2	2	2	2	2	2	2	2	2	3, 5	4	1	

2. Transmit event, tx

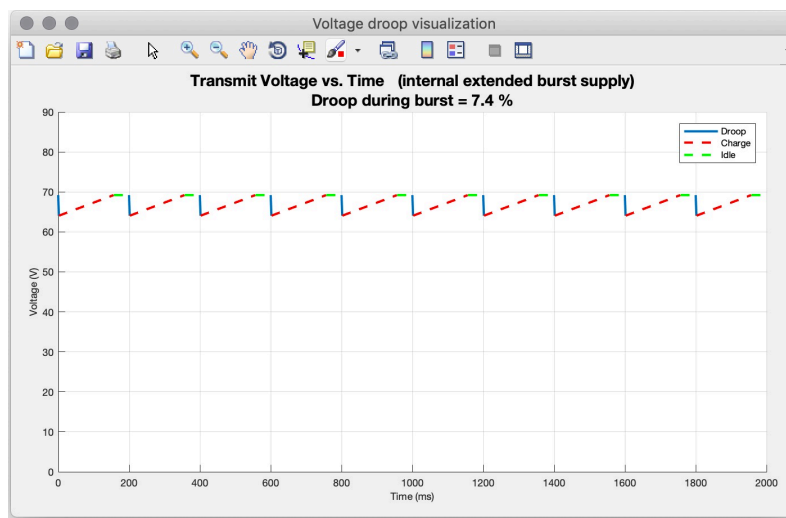
Below is the display of TX(24) after running SetUpL11_5vWideBeam. The Transducer, TX, and TW structures are displayed on the left, as well as the TX.Apod and TX.Delay are plotted on the right figure. In addition, two push button are available to plot the transmit waveform and TXPD which is the transmit beam profile.



It also provides the 3D view for a 2D array, as shown below, the user is able to rotate the 3D figure of TX.Apod and TX.Delay to have better view angle.

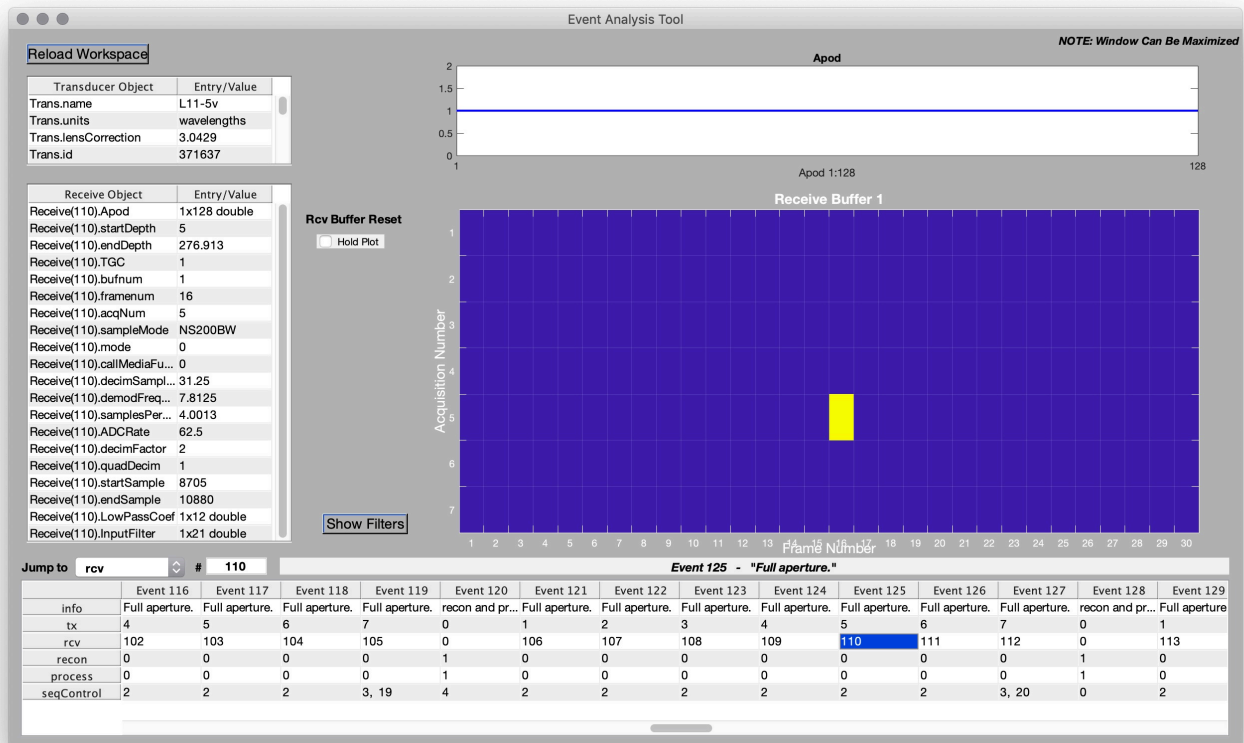


If the extend burst is used, a push button “Show Voltage Droop” will appear to display how the Voltage will droop in the entire event sequence. Please refer to the Appendix in the HIFUPlex User Documentation in our web portal, <http://verasonicscommunity.com/>, about the details.



3. Receive event, rcv

If any rcv event is selected, all attributes of the Receive will be displayed on the left and the illustration of the memory allocation will be on the right, with the Receive.Apod figure on the top of it. Below is the example of Receive(110) after running SetUpL11_5vWideBeam.



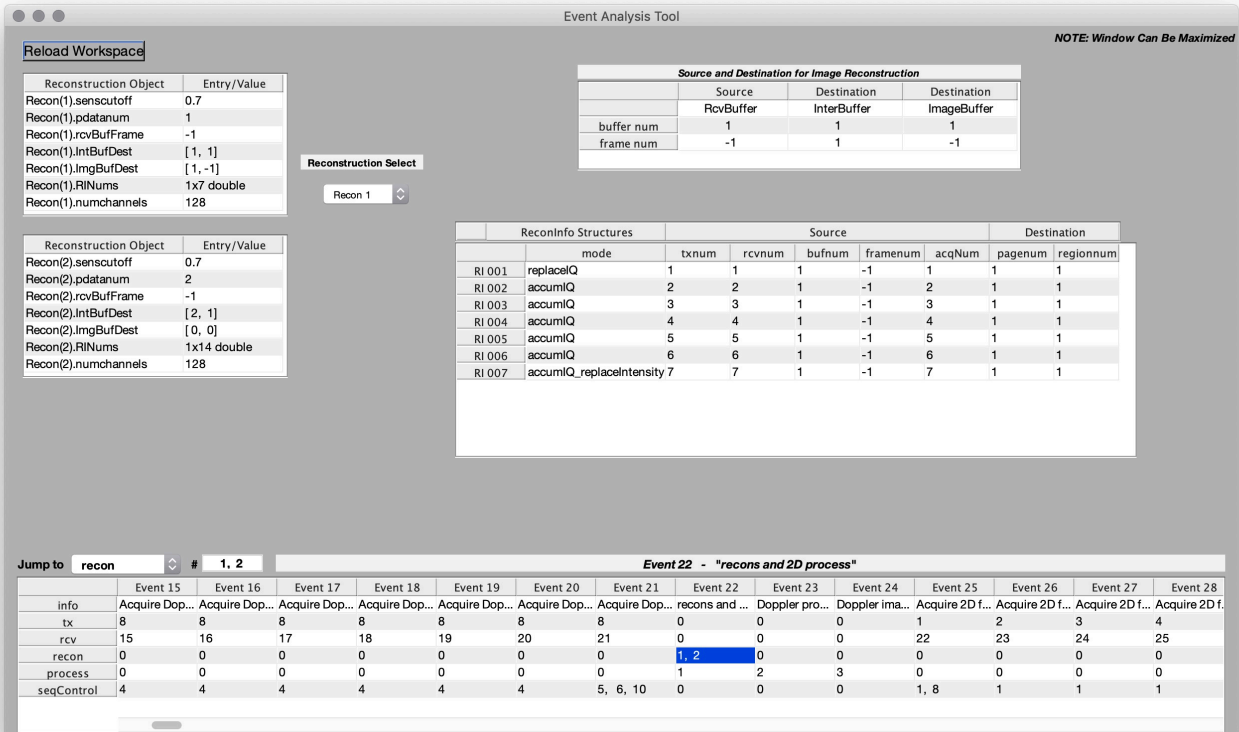
If initializeOnly is used, all Receive attributes are listed on the left. Many important attributes are provided by initializeOnly feature, including the demodFreq, ADCRate, startSample, and endSample. The endSample will be used for frame size calculation which is necessary for DMA transfer rate, so it's recommended to use initializeOnly feature unless the Matlab crashes during the initialization process.

If the "Hold Plot" check box is checked, the yellow area will not be cleared when another rcv is selected. It illustrates how the continuous RcvData is stored. Please note that it's easier to imagine the memory allocation with a 2-D display but the RcvData is actually three dimensional. The first one is depth along the acquisition numbers, the second one is number of elements, and the third one is the frame number. Please refer to our online training videos for more details: <http://verasonicscommunity.com/>

In addition, the hardware filter of each rcv event is independent, so the "Show Filters" button can be used to call the filterTool to display the filter setting of current rcv number.

4. Reconstruction event, recon

Usually, the Recon and ReconInfo is complicated and, if the script crashes Matlab, you should use EventAnalysisTool to inspect the Recon/ReconInfo setting. For instance, after running SetUpL11_5vFlashDoppler and execute EventAnalysisTool, if the “Jump to” feature is set to recon 1, the result is shown below:



The screenshot displays the Event Analysis Tool interface. At the top, there's a 'Reload Workspace' button and a 'Reconstruction Select' dropdown menu set to 'Recon 1'. The main area is divided into several sections:

- Reconstruction Object Entry/Value:** A table listing parameters for Recon(1) and Recon(2).
- Source and Destination for Image Reconstruction:** A table showing Source (RcvBuffer, InterBuffer, ImageBuffer) and Destination (buffer num, frame num) for Recon(1).
- ReconInfo Structures:** A table listing ReconInfo structures (RI 001 to RI 007) with their respective mode, txnum, rcvnum, bufnum, framenum, acqNum, pagenum, and regionnum.
- Jump to:** A dropdown menu set to 'recon' and a button labeled '# 1, 2'.
- Event 22 - "recons and 2D process":** A table showing event data for Event 15 to Event 28, including info, tx, rcv, recon, process, and seqControl.

In our software, multiple Recons can be included in the same event and the first index will be displayed as default. The Recon definition is listed on the left and the ReconInfo and associated Receive attributes, including bufnum, framenum, and acqNum, are listed on the bottom right. The top right table indicates the source and destination of Image Reconstruction. It helps use to understand possible errors regarding the source and destination in the Recon settings. More information can be found in the training manual and tutorial videos.

The drop-down menu is used to selected different Recon numbers when multiple recons are used in the same event. If the user select Recon 2, the table of ReconInfo will be changed, as shown in the next page:

Event Analysis Tool

NOTE: Window Can Be Maximized

Reload Workspace

Reconstruction Object	Entry/Value
Recon(1).senscutoff	0.7
Recon(1).pdatanum	1
Recon(1).rcvBufFrame	-1
Recon(1).IntBufDest	[1, 1]
Recon(1).ImgBufDest	[1, -1]
Recon(1).RINums	1x7 double
Recon(1).numchannels	128

Reconstruction Select

Recon 2

Source and Destination for Image Reconstruction			
	Source	Destination	Destination
	RcvBuffer	InterBuffer	ImageBuffer
buffer num	1	2	0
frame num	-1	1	0

ReconInfo Structures		Source				Destination	
	mode	txnum	rcvnum	bufnum	framenum	acqNum	pagenum regionnum
RI 008	replacelQ	8	8	1	-1	8	1 1
RI 009	replacelQ	8	9	1	-1	9	2 1
RI 010	replacelQ	8	10	1	-1	10	3 1
RI 011	replacelQ	8	11	1	-1	11	4 1
RI 012	replacelQ	8	12	1	-1	12	5 1
RI 013	replacelQ	8	13	1	-1	13	6 1
RI 014	replacelQ	8	14	1	-1	14	7 1
RI 015	replacelQ	8	15	1	-1	15	8 1
RI 016	replacelQ	8	16	1	-1	16	9 1
RI 017	replacelQ	8	17	1	-1	17	10 1
RI 018	replacelQ	8	18	1	-1	18	11 1

Jump to recon # 1, 2

Event 22 - "recons and 2D process"

	Event 15	Event 16	Event 17	Event 18	Event 19	Event 20	Event 21	Event 22	Event 23	Event 24	Event 25	Event 26	Event 27	Event 28
info	Acquire Dop...	Acquire Dop...	Acquire Dop...	Acquire Dop...	Acquire Dop...	Acquire Dop...	Acquire Dop...	recons and ...	Doppler pro...	Doppler ima...	Acquire 2D f...	Acquire 2D f...	Acquire 2D f...	Acquire 2D f...
tx	8	8	8	8	8	8	8	0	0	0	1	2	3	4
rcv	15	16	17	18	19	20	21	0	0	0	22	23	24	25
recon	0	0	0	0	0	0	0	1, 2	0	0	0	0	0	0
process	0	0	0	0	0	0	0	1	2	3	0	0	0	0
seqControl	4	4	4	4	4	4	5, 6, 10	0	0	0	1, 8	1	1	1

5. Process event, process

The EventAnalysisTool only displays all attributes in the selected Process number.

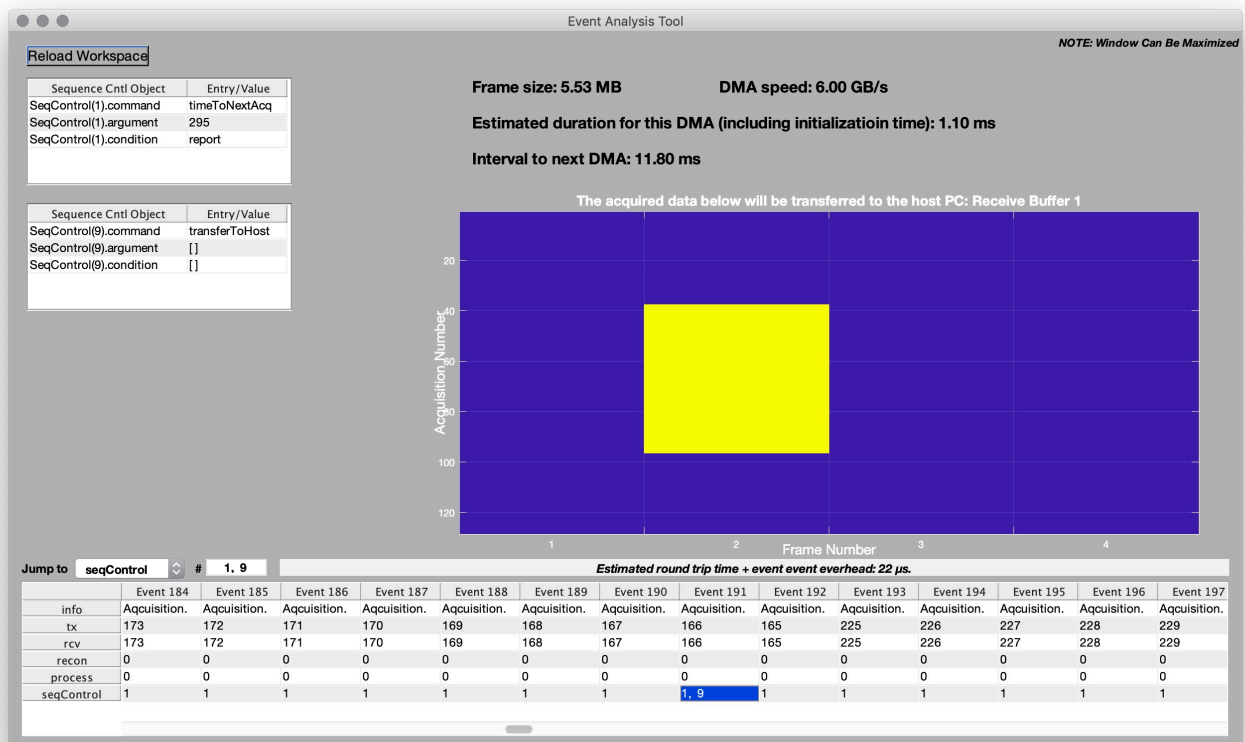
6. Flow control, seqControl

Most of the time, the EventAnalysisTool displays all attributes in one or multiple SeqControl objects shown in the same event. If "timeToNextAcq" is defined, the EventAnalysisTool will estimate the interval required between consecutive Transmit/Receive events. If the estimated interval is larger than the value defined for timeToNextAcq, a warning information will be displayed in the gray area which shows the event information, like:

Estimated round trip time + event overhead: 74 μ s. A timeToNextAcq warning might be displayed

Actually, a similar warning would appear when the EventAnalysisTool starts.

If the selected seqControl contains the “transferToHost” command, the estimated frame size, DMA speed, and transfer rate will be displayed as below. If the timeToNextAcq is defined in the same event, the interval to next transferToHost will be calculated as well, and a timeToNextAcq warning might be displayed because next DMA will not be initialized before the completion of current DMA. The DMA speed is system dependent and the transfer rate is estimated, so all information is provided as reference. If any bug is discovered, please email to Verasonics support (support@verasonics.com).



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Verasonics, Inc.
11335 NE 122nd Way, Suite 100,
Kirkland, WA 98034
USA